

PART 1 — IP PHONES

1. Traditional Phone vs IP Phone

Traditional Phone

Traditional telephone systems use the:

- **PSTN** → Public Switched Telephone Network
- Also called **POTS** → Plain Old Telephone Service

Voice traffic traditionally used a separate telephone network.

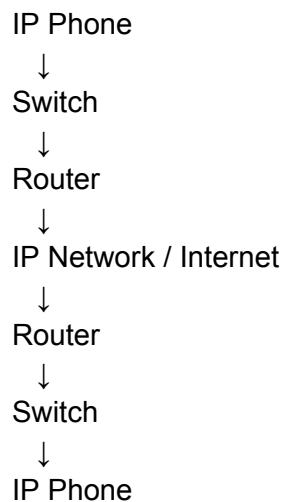
IP Phone

An IP phone uses:

VoIP — Voice over IP

Voice is converted into data and transmitted through an **IP network**.

Example:



Important

An IP phone is essentially an **end host** connected to a switch.

It participates in an IP network just like a PC does.

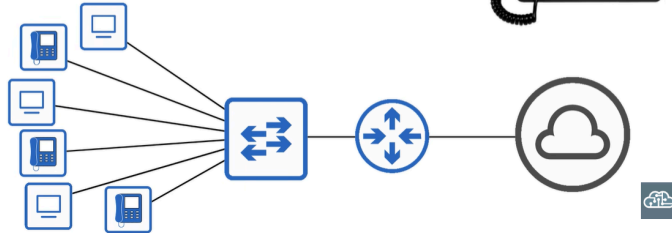


Image Placement 1



IP Phones

- Traditional phones operate over the *public switched telephone network* (PSTN).
- Sometimes this is called POTS (Plain Old Telephone Service).
- IP phones use VoIP (Voice over IP) technologies to enable phone calls over an IP network, such as the Internet.
- IP phones are connected to a switch just like any other end host.



Use the image showing:

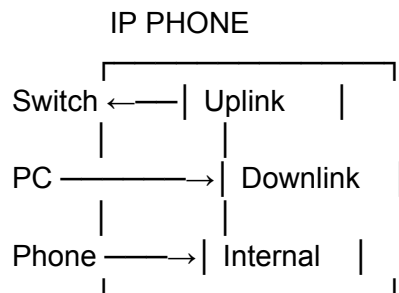
IP Phones → Switch → Router → Cloud
 PCs → Switch

PART 2 — IP PHONE INTERNAL SWITCH

2. IP Phone has a 3-Port Internal Switch

An IP phone commonly contains a small **internal 3-port switch**.

The three connections are:



Ports

1. **Uplink**
 - Connects to the external network switch.
2. **Downlink**

- Connects to the user's PC.
- 3. **Internal connection**
 - Connects to the phone itself.

Therefore:

The PC and IP phone can share **one physical switch port**.

PART 3 — VOICE VLAN

3. Why use a Voice VLAN?

Suppose we have:

PC — IP Phone — Switch

Both PC and phone are using the same physical cable.

But we want to logically separate:

PC traffic → Data VLAN
IP Phone traffic → Voice VLAN

Example:

VLAN 10 → Data
VLAN 11 → Voice

Why?

Because voice traffic has different requirements from normal data traffic.

Voice is sensitive to:

- Delay
- Jitter
- Packet loss

So separating voice traffic into a **Voice VLAN** helps network administrators manage it separately and apply appropriate QoS policies.

4. Access VLAN + Voice VLAN

Example Cisco configuration:

```
interface gigabitethernet0/0
switchport mode access
switchport access vlan 10
switchport voice vlan 11
```

This means:

PC traffic → VLAN 10
Phone traffic → VLAN 11

Very important

This interface is still considered an **ACCESS PORT**.

It is **NOT a trunk port**, even though traffic from two VLANs is using it.

Why?

Because the switchport is configured as:

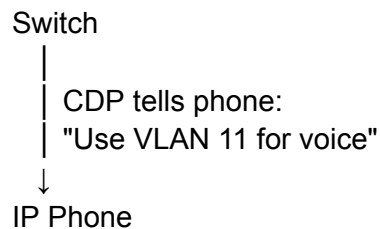
```
switchport mode access
```

5. How does the phone know to tag its traffic?

The switch tells the IP phone which VLAN to use.

Cisco commonly uses **CDP** for this purpose.

Conceptually:



Then:

PC traffic

PC → Switch

Untagged

Phone traffic

Phone → Switch

802.1Q tagged

VLAN ID = 11

So the same physical link carries:

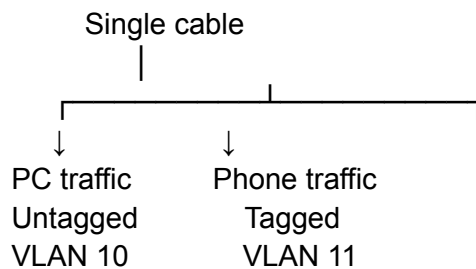
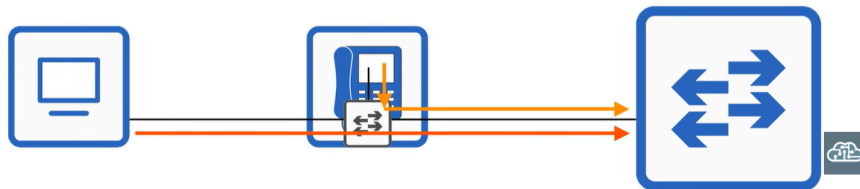


Image Placement 2



IP Phones

- IP phones have an internal 3-port switch.
 - 1 port is the 'uplink' to the external switch.
 - 1 port is the 'downlink' to the PC.
 - 1 port connects internally to the phone itself.
- This allows the PC and the IP phone to share a single switch port. Traffic from the PC passes through the IP phone to the switch.
- It is recommended to separate 'voice' traffic (from the IP phone) and 'data' traffic (from the PC) by placing them in separate VLANs.
 - This can be accomplished using a *voice VLAN*
 - Traffic from the PC will be untagged, but traffic from the phone will be tagged with a VLAN ID



PC1 → PH1 → SW1

VLAN 11

6. Voice VLAN Configuration

```
SW1(config)# interface gigabitethernet0/0
```

```
SW1(config-if)# switchport mode access
```

```
SW1(config-if)# switchport access vlan 10
SW1(config-if)# switchport voice vlan 11
```

Meaning

Traffic	VLAN	Tagging
PC/Data	10	Untagged
IP Phone/Voice	11	Tagged

Important exam point

Even though two VLANs are carried:

The interface is still an access port, not a trunk port.

You may see:

Administrative Mode: static access

Operational Mode: static access

Access Mode VLAN: 10

Voice VLAN: 11

That confirms it is an **access port** with a configured voice VLAN.

Image Placement 3

Place the **show interfaces g0/0 switchport output** here.

JEREMY'S NETWORK
IP Phones / Voice VLAN

```

SW1(config)#interface gigabitethernet0/0
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#switchport voice vlan 11

SW1#show interfaces g0/0 switchport
Name: Gi0/0
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: negoti
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (VLAN0010)
Trunking Native Mode VLAN: 1 (default)
Administrative Native VLAN tagging: enabled
Voice VLAN: 11 (VLAN0011)
! [output omitted]

```

PC1

PH1

SW1

Diagram showing PC1 connected to PH1 (IP Phone) and PH1 connected to SW1 (Switch). The connection between PH1 and SW1 is labeled 'VLAN11' and 'g0/0'.

PC1 will send traffic untagged, as normal. SW1 will use CDP to tell PH1 to tag PH1's traffic in VLAN 11.

Although the interface sends/receives traffic from two VLANs, it is not considered a trunk port. It is considered an access port.

PART 4 — TRUNK CONFUSION

You may see:

show interfaces g0/0 trunk

and notice:

Status: not-trunking

That's completely normal.

Even though:

VLAN 10 → data

VLAN 11 → voice

are present on the port, the port itself is not operating as a normal 802.1Q trunk.

Remember:

Voice VLAN ≠ trunk port.

Image Placement 4

Place the **show interfaces trunk** output showing **not-trunking** here.



IP Phones / Voice VLAN

```
SW1#show interfaces trunk
SW1#
SW1#show interfaces g0/0 trunk

Port      Mode      Encapsulation  Status  Native vlan
Gi0/0     off       negotiate      not-trunking  1

Port      Vlans allowed on trunk
Gi0/0     10-11

Port      Vlans allowed and active in management domain
Gi0/0     10-11

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10-11
```



PART 5 — POWER OVER ETHERNET

7. What is PoE?

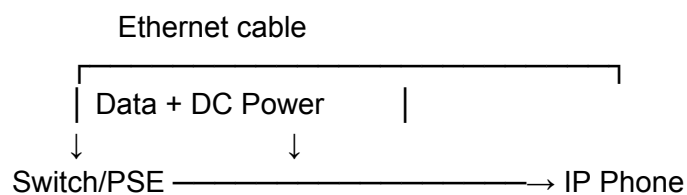
PoE = Power over Ethernet

PoE allows a network cable to carry:

- Data
- Electrical power

at the same time.

Example:



Without PoE:

Ethernet cable → Data

Power cable → Electricity

With PoE:

Ethernet cable → Data + Power

8. PSE and PD

Two important terms:

PSE

Power Sourcing Equipment

Usually:

PoE-enabled switch

It provides power.

PD

Powered Device

Examples:

- IP phone
- IP camera
- Wireless Access Point

It receives power.

PSE PD
Switch ——— Ethernet ———→ IP Phone
Data + Power

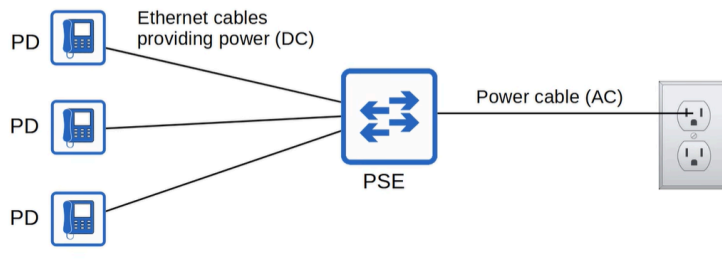


Image Placement 5



Power over Ethernet (PoE)

- PoE allows Power Sourcing Equipment (PSE) to provide power to Powered Devices (PD) over an Ethernet cable.
- Typically the PSE is a switch and the PDs are IP phones, IP cameras, wireless access points, etc.
- The PSE receives AC power from the outlet, converts it to DC power, and supplies that DC power to the PDs.



9. How PoE works

The switch doesn't blindly send large amounts of electricity.

It first determines whether the connected device requires power.

Simplified process:

1. Device connected
- ↓
2. PSE detects device
- ↓
3. PSE determines power requirement
- ↓
4. PSE supplies appropriate power
- ↓
5. PSE continues monitoring

Why?

Because:

Too much electrical current can damage a device.

10. PoE Power Policing

PoE can also have **power policing**.

The switch can prevent a powered device from consuming excessive power.

Example:

IP Phone
↓
Requests/draws power
↓
Switch monitors power
↓
Too much?
↓
Take configured action

power inline police

```
interface g0/0  
power inline police
```

Default action is effectively:

power inline police action err-disable

If the device draws excessive power:

Port → err-disabled
Syslog message

The port can be recovered using:

```
shutdown  
no shutdown
```

11. power inline police action log

```
interface g0/0  
power inline police action log
```

If excessive power is detected:

- Log the event
- Restart the interface/power

- Do not permanently put the interface into err-disabled state

Compare

Command	Excessive power
<code>power inline police</code>	Err-disable port
<code>power inline police action log</code>	Log event/restart without err-disable

Image Placement 6



Power over Ethernet (PoE)

- Too much electrical current can damage electrical devices.
- PoE has a process to determine if a connected device needs power, and how much power it needs.
 - When a device is connected to a PoE-enabled port, the PSE (switch) sends low power signals, monitors the response, and determines how much power the PD needs.
 - If the device needs power, the PSE supplies the power to allow the PD to boot.
 - The PSE continues to monitor the PD and supply the required amount of power (but not too much!)
- *Power policing* can be configured to prevent a PD from taking too much power.
 - `power inline police` configures power policing with the default settings: disable the port and send a Syslog message if a PD draws too much power.
 - equivalent to `power inline police action err-disable`
 - the interface will be put in an 'error-disabled' state and can be re-enabled with `shutdown` followed by `no shutdown`.
 - `power inline police action log` does not shut down the interface if the PD draws too much power. It will restart the interface and send a Syslog message.

12. PoE Standards

Type	Standard	Power	Wire pairs
Cisco Inline Power	Cisco proprietary	~7 W	2
PoE Type 1	802.3af	15 W	2
PoE+ Type 2	802.3at	30 W	2
UPOE Type 3	802.3bt	60 W	4
UPOE+ Type 4	802.3bt	100 W	4

Memory

802.3af → PoE

802.3at → PoE+

802.3bt → Type 3 / Type 4

Image Placement 7

 **Power over Ethernet (PoE)**

```
SW1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)# int g0/0
SW1(config-if)# power inline police
SW1(config-if)# end
SW1# show power inline police g0/0
Available:800(w) Used:32(w) Remaining:768(w)
Interface Admin Oper Admin Oper Cutoff Oper
          State State Police Police Power Power
-----
Gi2/1     auto  on      errdisable ok      17.2  16.7
```

 PC1  PH1  SW1

 **Power over Ethernet (PoE)**

Name	Standard #	Watts	Powered Wire Pairs
Cisco Inline Power (ILP)	Made by Cisco, not standard	7	2
PoE (Type 1)	802.3af	15	2
PoE+ (Type 2)	802.3at	30	2
UPoE (Type 3)	802.3bt	60	4
UPoE+ (Type 4)	802.3bt	100	4

PART 6 — WHY QoS?

Now we reach the important part.

13. Old Networks

Originally, voice and data often used separate networks.

Voice

↓
PSTN

Data
↓
IP Network

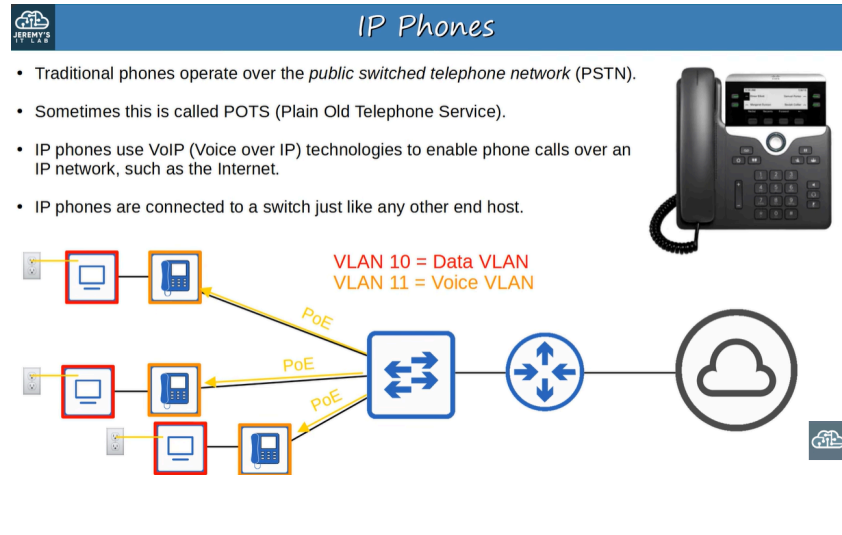
They didn't compete for the same bandwidth.

Therefore:

QoS wasn't as important.

Image Placement 8

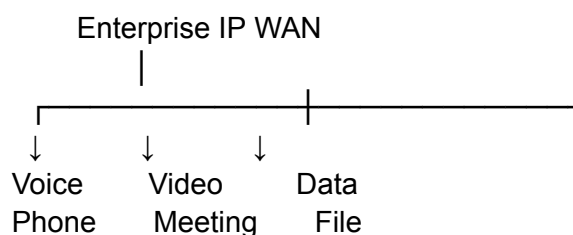
Place the **old PSTN vs Enterprise IP WAN** diagram here.



14. Modern Converged Networks

Modern networks combine different traffic types on the same IP network.

For example:



Now they compete for bandwidth.

For example:

Voice
Video
Web
FTP
File transfers
↓
SAME LINK

This creates congestion.

Therefore:

QoS is needed to apply different treatment to different types of traffic.

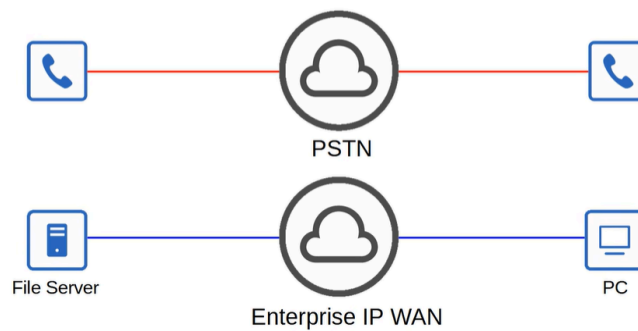
Image Placement 9

Place the **modern converged network QoS diagram** here.



Quality of Service (QoS)

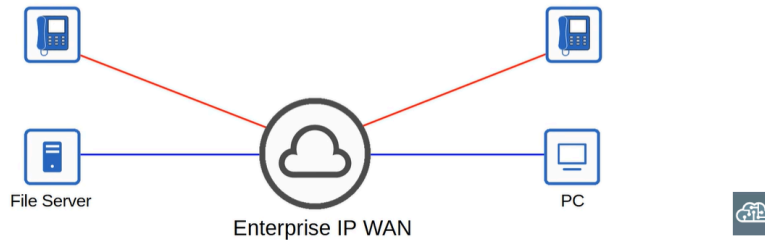
- Voice traffic and data traffic used to use entirely separate networks.
 - **Voice traffic** used the PSTN
 - **Data traffic** used the IP network (enterprise WAN, Internet, etc)
- QoS wasn't necessary as the different kinds of traffic didn't compete for bandwidth.





Quality of Service (QoS)

- Modern networks are typically *converged networks* in which IP phones, video traffic, regular data traffic, etc all share the same IP network.
- This enable cost savings as well as more advanced features for voice and video traffic, for example integrations with collaboration software (Cisco WebEx, Microsoft Teams, etc).
- However, the different kinds of traffic now have to compete for bandwidth.
- QoS is a set of tools used by network devices to apply different treatment to different packets.



PART 7 — WHAT DOES QoS ACTUALLY MANAGE?

There are four major characteristics you should know:

15. Bandwidth

Bandwidth = capacity of a link.

Example:

100 Mbps

1 Gbps

10 Gbps

QoS can reserve a portion of available bandwidth for particular traffic.

Example:

100 Mbps link

Voice → 20%

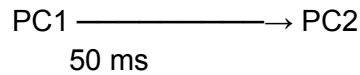
Data A → 30%

Other → 50%

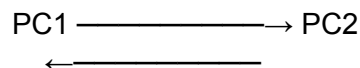
16. Delay

Delay = time taken for traffic to travel.

One-way delay



Two-way delay



It represents the time for traffic to go there and return.

Image Placement 10

Place the **one-way vs two-way delay diagram** here.



Quality of Service (QoS)

- QoS is used to manage the following characteristics of network traffic:
 - Bandwidth**
 - The overall capacity of the link, measured in bits per second (Kbps, Mbps, Gbps, etc)
 - QoS tools allow you to reserve a certain amount of a link's bandwidth for specific kinds of traffic.
For example: 20% voice traffic, 30% for specific kinds of data traffic, leaving 50% for all other traffic.
 - Delay**
 - The amount of time it takes traffic to go from source to destination = **one-way delay**
 - The amount of time it takes traffic to go from source to destination and return = **two-way delay**

The diagram illustrates network delay between two phones, PH1 and PH2, connected via a cloud representing the network. A red arrow points from PH1 to PH2, labeled 'one-way delay'. Two orange arrows point from PH2 back to PH1, labeled 'two-way delay'.



17. Jitter

This is extremely important for voice.

Jitter

Variation in packet delay.

Suppose packets arrive:

Packet 1 → 20 ms

Packet 2 → 25 ms

Packet 3 → 80 ms

Packet 4 → 30 ms

The delay isn't consistent.

That's jitter.

Voice doesn't like this.

So IP phones can use a:

Jitter buffer

The jitter buffer temporarily holds audio packets and helps provide smoother playback.



Image Placement 11

Place the **Jitter + Loss** slide here.



Quality of Service (QoS)

- QoS is used to manage the following characteristics of network traffic:
 - 1) **Bandwidth**
 - The overall capacity of the link, measured in bits per second (Kbps, Mbps, Gbps, etc)
 - QoS tools allow you to reserve a certain amount of a link's bandwidth for specific kinds of traffic.
For example: 20% voice traffic, 30% for specific kinds of data traffic, leaving 50% for all other traffic.
 - 2) **Delay**
 - The amount of time it takes traffic to go from source to destination = **one-way delay**
 - The amount of time it takes traffic to go from source to destination and return = **two-way delay**
 - 3) **Jitter**
 - The variation in one-way delay between packets sent by the same application
 - IP phones have a 'jitter buffer' to provide a fixed delay to audio packets.
 - 4) **Loss**
 - The % of packets sent that do not reach their destination
 - Can be caused by faulty cables.
 - Can also be caused when a device's packet *queues* get full and the device starts discarding packets.



18. Packet Loss

Packet loss means:

Packets were sent but did not reach the destination.

Causes include:

- Faulty cable
- Interface problems
- Congestion
- Full packet queue

For congestion:

Traffic arrives



Queue fills



Queue full



Packets discarded



Packet loss

19. Recommended Interactive Audio Targets

For acceptable interactive audio quality:

Metric	Recommended
One-way delay	≤ 150 ms
Jitter	≤ 30 ms
Loss	$\leq 1\%$

If these values become too high, call quality can noticeably degrade.

Image Placement 12

Place the **audio-quality standards** slide here.

PART 8 — QUEUING

Now we reach the part you were asking about earlier.

20. Why does a queue exist?

Suppose a router receives packets faster than it can transmit them through an outgoing interface.

Example:

Incoming traffic
↓
100 Mbps
↓
Router
↓
10 Mbps outgoing link

The router cannot immediately send everything.

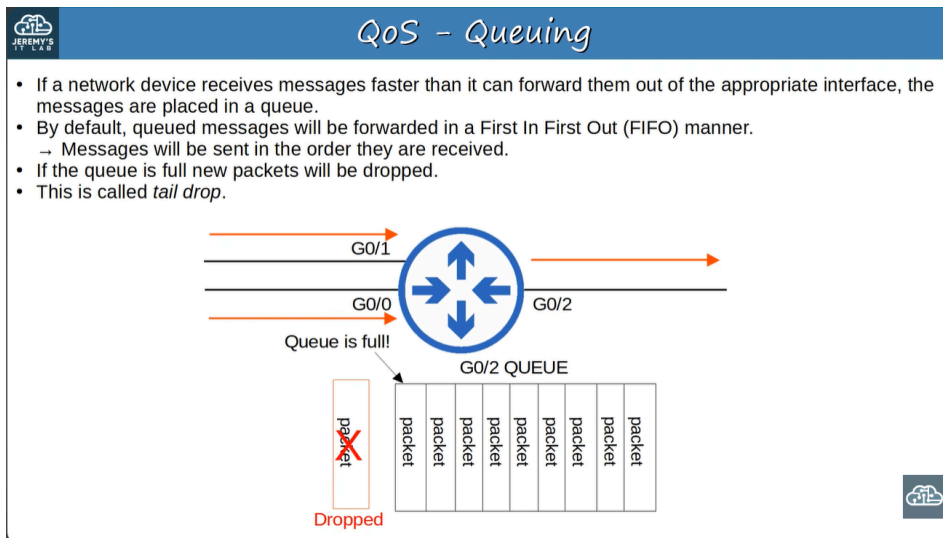
Therefore:

Packets
↓
QUEUE
↓
Outgoing interface



Image Placement 13

Place the **G0/1 + G0/0 → G0/2 queue diagram** here.



21. FIFO — First In, First Out

By default, queued packets are sent in the order they arrived.

Arrival:

A → B → C → D

Transmission:

A → B → C → D

This is:

FIFO

Simple, but FIFO does not understand traffic importance.

22. Tail Drop

What happens if the queue becomes full?

Suppose:

Queue capacity = 8 packets

Already:

P1 P2 P3 P4 P5 P6 P7 P8

Now P9 arrives.

There is no space.

Therefore:

P9 →  DROP

This is called:

Tail Drop

The arriving packet is discarded because the queue is full.

Important connection to your previous doubt

The router **does not tell TCP**:

"I dropped your packet."

It simply drops it.

TCP later discovers the loss through:

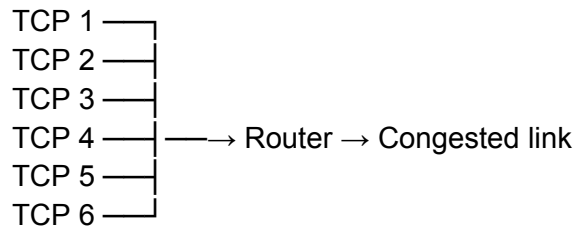
- Missing/duplicate ACKs
- Retransmission timeout

PART 9 — WHY IS TAIL DROP A PROBLEM?

23. TCP Global Synchronization

This is one of the most important concepts in these slides.

Suppose many TCP connections are using the same router.



The queue fills.

QUEUE FULL



TAIL DROP



Several TCP flows lose packets

Each affected TCP sender detects loss.

Therefore:

TCP 1 → decrease sending rate

TCP 2 → decrease sending rate

TCP 3 → decrease sending rate

TCP 4 → decrease sending rate

...

So many TCP flows reduce their sending rates at approximately the same time.

That's:

TCP Global Synchronization

24. What happens next?

After reducing their rates, the network becomes underutilized.

Congestion



Tail Drop



TCP windows decrease



Traffic decreases



Network becomes underutilized



TCP windows increase again



Traffic increases



Congestion returns

So you get a repeating cycle:

Congestion



Tail Drop



TCP windows decrease



Network underutilized



TCP windows increase



Congestion



...

This is why tail drop can produce inefficient utilization.

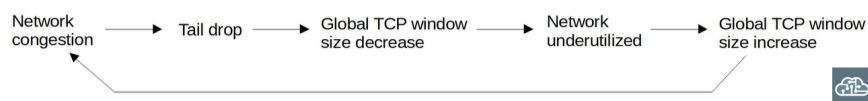


Image Placement 14



QoS - Queuing

- **Tail drop** is harmful because it can lead to **TCP global synchronization**.
- Review of the **TCP sliding window**:
 - Hosts using TCP use the 'sliding window' increase/decrease the rate at which they send traffic as needed.
 - When a packet is dropped it will be re-transmitted.
 - When a drop occurs, the sender will reduce the rate it sends traffic.
 - It will then gradually increase the rate again.
- When the queue fills up and **tail drop** occurs, all TCP hosts sending traffic will slow down the rate at which they send traffic.
- They will all then increase the rate at which they send traffic, which rapidly leads to more congestion, dropped packets, and the process repeats again.



PART 10 — RED

25. Random Early Detection

The solution introduced in your lecture is:

RED = Random Early Detection

The idea is very different from waiting until the queue is completely full.

Tail Drop

Queue:

 FULL

New packet



DROP

RED

RED starts taking action **before the queue becomes completely full**.

Conceptually:

Queue utilization

0% ————— 100%

↑

Too late

RED establishes thresholds.

When the queue reaches a certain level:

The router starts randomly dropping packets from selected TCP flows.

26. Why randomly drop packets?

Suppose:

TCP A
TCP B
TCP C
TCP D
TCP E

Instead of allowing the queue to fill completely and then dropping many packets together:

RED can cause selected flows to detect congestion at different times.

Example:

TCP A → packet dropped
TCP B → continues
TCP C → continues
TCP D → packet dropped
TCP E → continues

Therefore:

A ↓
D ↓

B —
C —
E —

Not everybody decreases their rate at exactly the same time.

This helps avoid:

TCP global synchronization

PART 11 — WRED

27. Weighted Random Early Detection

Standard RED treats traffic similarly.

But what if you want different treatment for different traffic classes?

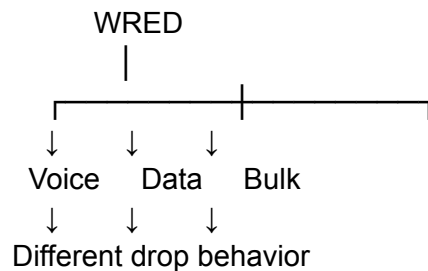
That's where:

WRED = Weighted Random Early Detection

comes in.

The "Weighted" part means packet-dropping behavior can depend on things such as **traffic class / packet marking**.

Conceptually:



This allows the administrator to control **which traffic is more likely to be dropped** when congestion begins.

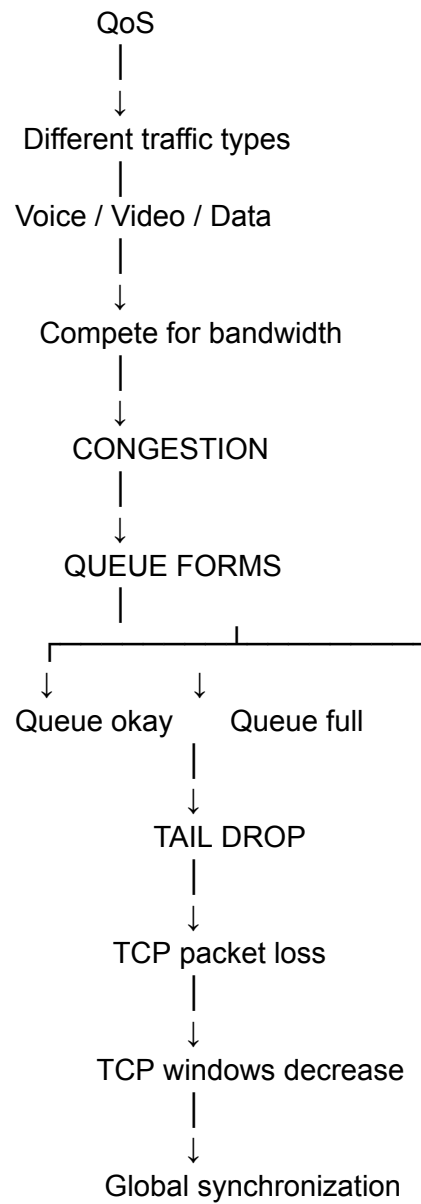
28. Tail Drop vs RED vs WRED

Feature	Tail Drop	RED	WRED
Wait until queue full?	Yes	No	No
Early dropping?	✗	✓	✓
Random dropping?	✗	✓	✓
Traffic classes considered?	✗	Generally no	✓
Helps global synchronization?	✗	✓	✓
More control over different traffic?	Low	Medium	High

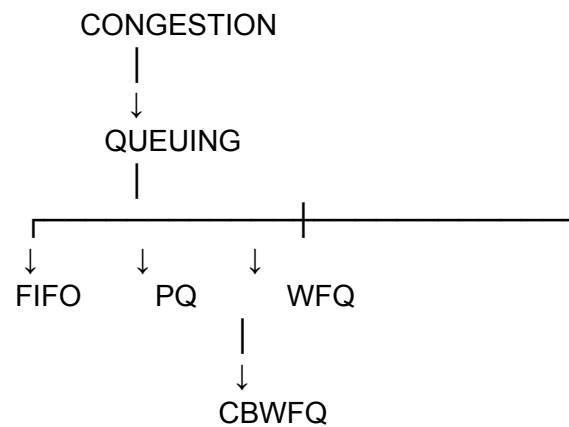


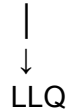
THE BIG PICTURE

This is the part I want you to memorize conceptually:

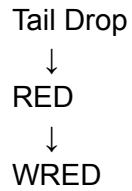


QoS introduces smarter mechanisms:





And for congestion avoidance:



1. QoS — Basic Idea

QoS (Quality of Service) gives certain traffic **priority over other traffic during congestion**.

Example:

Voice call + file download → during congestion, voice should get better treatment.

QoS flow

Classification → Marking → Queuing → Scheduling → Transmission

And for rate control:

Shaping / Policing

2. Classification

What is classification?

Classification = identifying what type of traffic a packet belongs to.

It puts packets into **traffic classes**.

Example:

Traffic	Class
---------	-------

Voice	High priority
Video	High priority
Web	Normal
File download	Low

How can traffic be classified?

Common methods:

- **ACL** → identifies traffic based on IP/port/etc.
- **NBAR** → Deep Packet Inspection, can identify applications up to Layer 7.
- **Layer 2/3 fields** → fields in Ethernet/IP headers.

Important idea

Before giving priority, the router must know what traffic it is.

3. Marking

After classification, we can **mark** the packet.

What is marking?

Adding a value to the packet that tells other network devices:

"This packet belongs to this traffic class."

Two important marking fields:

Layer 2 → PCP

PCP (Priority Code Point)

- Located in **802.1Q VLAN tag**
- **3 bits**
- 8 values → **0–7**
- Also called **CoS (Class of Service)**

Layer 3 → DSCP

DSCP (Differentiated Services Code Point)

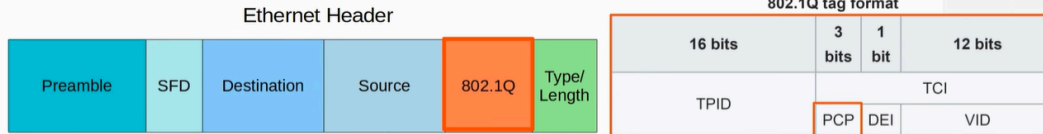
- Located in the **IP header**
- **6 bits**
- 64 values → **0–63**

Remember

PCP = Layer 2

DSCP = Layer 3

4. PCP / CoS



- PCP is also known as CoS (Class of Service). Its use is defined by IEEE 802.1p.

- 3 bits = 8 possible values ($2^3 = 8$).

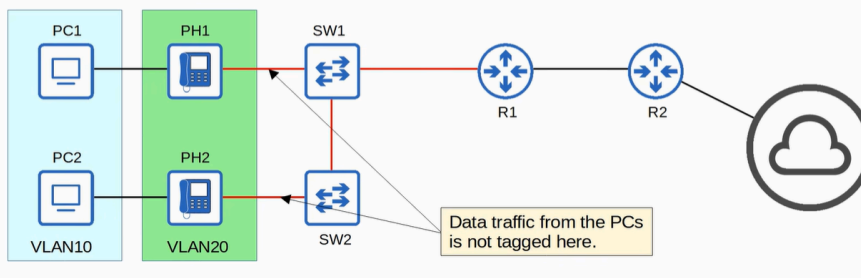
'Best effort' delivery means there is no guarantee that data is delivered or that it meets any QoS standard. This is regular traffic, not high-priority.

IP phones **mark** call signaling traffic (used to establish calls) as PCP3. They **mark** the actual voice traffic as PCP5.

PCP value	Traffic types
0	Best effort (default)
1	Background
2	Excellent effort
3	Critical applications
4	Video
5	Voice
6	Internetwork control
7	Network control



- Because PCP is found in the dot1q header, it can only be used over the following connections:
 - trunk links
 - access links with a voice VLAN
- In the diagram below, traffic between R1 and R2, or between R2 and external destinations will not have a dot1q tag. So, traffic over those links PCP cannot be marked with a PCP value.



PCP is inside the **802.1Q VLAN tag**.

802.1Q Tag

```

+-----+
| TPID | PCP | DEI | VID |
|  3bit 1bit 12 bit  |
+-----+
    
```

PCP values

PCP	Traffic
-----	---------

0	Best effort
---	-------------

- 1 Background
- 2 Excellent effort
- 3 Critical applications
- 4 Video
- 5 Voice
- 6 Internetwork control
- 7 Network control

Important

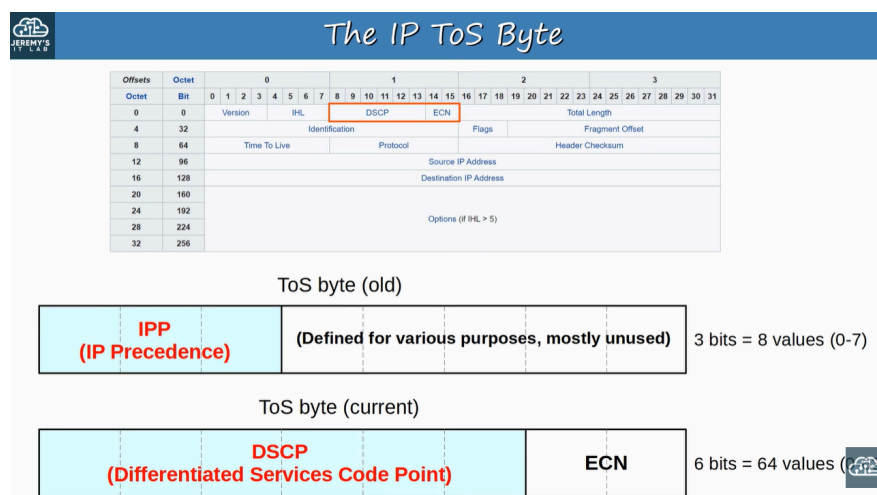
PCP requires an **802.1Q** tag.

Therefore, it can be used on:

- Trunk links
- Access ports with a **voice VLAN**

It cannot be used on a normal **untagged PC-to-router Layer-3** link.

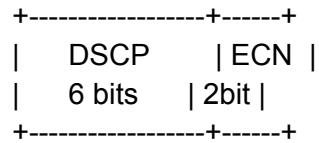
5. IP ToS Byte



The IP header contains the field historically called the **ToS (Type of Service) byte**.

Today, it is divided into:

8 bits



Old ToS

Previously:

IPP (IP Precedence) → 3 bits

Current

DSCP → 6 bits

ECN → 2 bits

So:

IPP → older QoS marking

DSCP → modern QoS marking

6. IP Precedence (IPP)



IP Precedence

IPP (IP Precedence)	(Defined for various purposes, mostly unused)	3 bits = 8 values (0-7)
--------------------------------	---	-------------------------

- Standard IPP markings are similar to PCP:
 - 6 and 7 are reserved for 'network control' traffic (ie. OSPF messages between routers)
 - 5 = voice
 - 4 = video
 - 3 = voice signaling
 - 0 = best effort
- With 6 and 7 reserved, 6 possible values remain.
- Although 6 values is sufficient for many networks, the QoS requirements of some networks demand more flexibility.



IPP uses **3 bits**.

Therefore:

$2^3 = 8$ values $\rightarrow$ 0–7

Common values:

IPP	Meaning
0	Best effort
3	Voice signaling
4	Video
5	Voice
6–7	Network control

IPP is mostly an **older marking method**.

7. DSCP



DSCP

DSCP (Differentiated Services Code Point)	ECN	6 bits = 64 values (0-63)
--	------------	---------------------------

- RFC 2474 (1998) defines the DSCP field, and other 'DiffServ' RFCs elaborate on its use.
- With IPP updated to DSCP, new standard markings had to be decided upon.
 - By having generally agreed upon standard markings for different kinds of traffic, QoS design & implementation is simplified, QoS works better between ISPs and enterprises, among other benefits.
- You should be aware of the following standard markings:
 - Default Forwarding (DF) – best effort traffic
 - Expedited Forwarding (EF) – low loss/latency/jitter traffic (usually voice)
 - Assured Forwarding (AF) – A set of 12 standard values
 - Class Selector (CS) – A set of 8 standard values, provides backward compatibility with IPP





DSCP

```

R1(config)#class-map TEST
R1(config-cmap)#match dscp ?
<0-63> Differentiated services codepoint value
af11 Match packets with AF11 dscp (001010)
af12 Match packets with AF12 dscp (001100)
af13 Match packets with AF13 dscp (001110)
af21 Match packets with AF21 dscp (010010)
af22 Match packets with AF22 dscp (010100)
af23 Match packets with AF23 dscp (010110)
af31 Match packets with AF31 dscp (011010)
af32 Match packets with AF32 dscp (011100)
af33 Match packets with AF33 dscp (011110)
af41 Match packets with AF41 dscp (100010)
af42 Match packets with AF42 dscp (100100)
af43 Match packets with AF43 dscp (100110)
cs1 Match packets with CS1(precedence 1) dscp (001000)
cs2 Match packets with CS2(precedence 2) dscp (010000)
cs3 Match packets with CS3(precedence 3) dscp (011000)
cs4 Match packets with CS4(precedence 4) dscp (100000)
cs5 Match packets with CS5(precedence 5) dscp (101000)
cs6 Match packets with CS6(precedence 6) dscp (110000)
cs7 Match packets with CS7(precedence 7) dscp (111000)
default Match packets with default dscp (000000)
ef Match packets with EF dscp (101110)
  
```



DSCP provides much more flexibility than IP Precedence.

6 bits = 64 possible values

DSCP = 6 bits

000000 → 0

111111 → 63

Important standard DSCP categories:



DF / EF

DF (Default Forwarding):

32	16	8	4	2	1		
0	0	0	0	0	0		

- DF is used for best-effort traffic.
- The DSCP marking for DF is 0.

1. DF — Default Forwarding

Best-effort traffic.

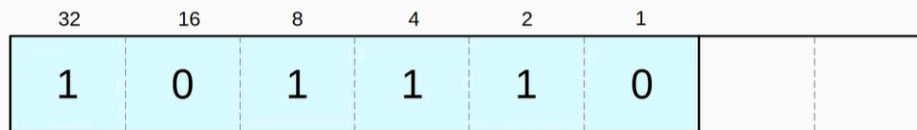
DSCP = 0

Think:

Normal traffic.

2. EF — Expedited Forwarding

EF (Expedited Forwarding):



- EF is used for traffic that requires low loss/latency/jitter.
- The DSCP marking for EF is 46.



Used for traffic requiring:

- Low delay
- Low jitter
- Low packet loss

Usually:

Voice

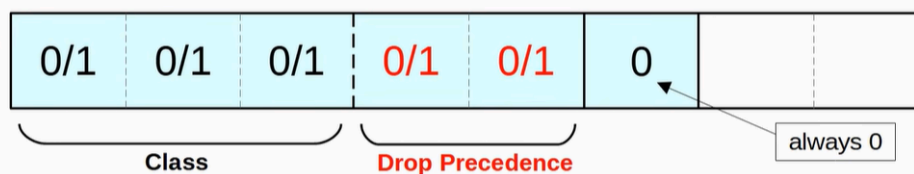
EF = DSCP 46

Binary:

46 = 101110

3. AF — Assured Forwarding

- AF (Assured Forwarding) defines four traffic classes. All packets in a class have the same priority.
- Within each class, there are three levels of *drop precedence*.
 - Higher drop precedence = more likely to drop the packet during congestion



= AFXY

AF provides:

- **4 classes**
- Each class has **3 drop precedences**

So:

4 × 3 = 12 AF values

Format:

AFX Y

Where:

- **X = traffic class**
- **Y = drop precedence**

Example:

AF11

= Class 1 + Drop Precedence 1

4. CS — Class Selector

CS

- **CS** (Class Selector) defines eight DSCP values for backward compatibility with IPP.
- The three bits that were added for DSCP are set to 0, and the original IPP bits are used to make 8 values.

	(32) 4	(16) 2	(8) 1	(4)	(2)		
	0/1	0/1	0/1	0	0	0	
IPP:	0	1	2	3	4	5	6 7
CS:	CS0	CS1	CS2	CS3	CS4	CS5	CS6 CS7
DSCP: (decimal)	0	8	16	24	32	40	48 56



Provides **8 values** for backward compatibility with IP Precedence.

CS0 → DSCP 0

CS1 → DSCP 8

CS2 → DSCP 16
CS3 → DSCP 24
CS4 → DSCP 32
CS5 → DSCP 40
CS6 → DSCP 48
CS7 → DSCP 56

Easy formula:

$$\text{CS number} \times 8 = \text{DSCP value}$$

Example:

$$\text{CS5} \rightarrow 5 \times 8 = 40$$

8. AF — Assured Forwarding

This is the part you need to understand carefully.

AF has:

4 classes

AF1
AF2
AF3
AF4

Within each class:

3 drop precedences

1 = lowest chance of dropping
2 = medium
3 = highest chance of dropping

So:

AF11 AF12 AF13
AF21 AF22 AF23
AF31 AF32 AF33
AF41 AF42 AF43

Example

AF41

- Class = 4
- Drop precedence = 1

AF43

- Class = 4
- Drop precedence = 3

Both are **Class 4**, but AF43 is more likely to be dropped during congestion.

AF table

	Drop 1	Drop 2	Drop 3
Class 1	AF11 = 10	AF12 = 12	AF13 = 14
Class 2	AF21 = 18	AF22 = 20	AF23 = 22
Class 3	AF31 = 26	AF32 = 28	AF33 = 30
Class 4	AF41 = 34	AF42 = 36	AF43 = 38

Most important concept

Class → priority category

Drop precedence → how likely the packet is to be dropped

9. RFC 4954 — Traffic Marking Recommendations

Important mappings to remember:

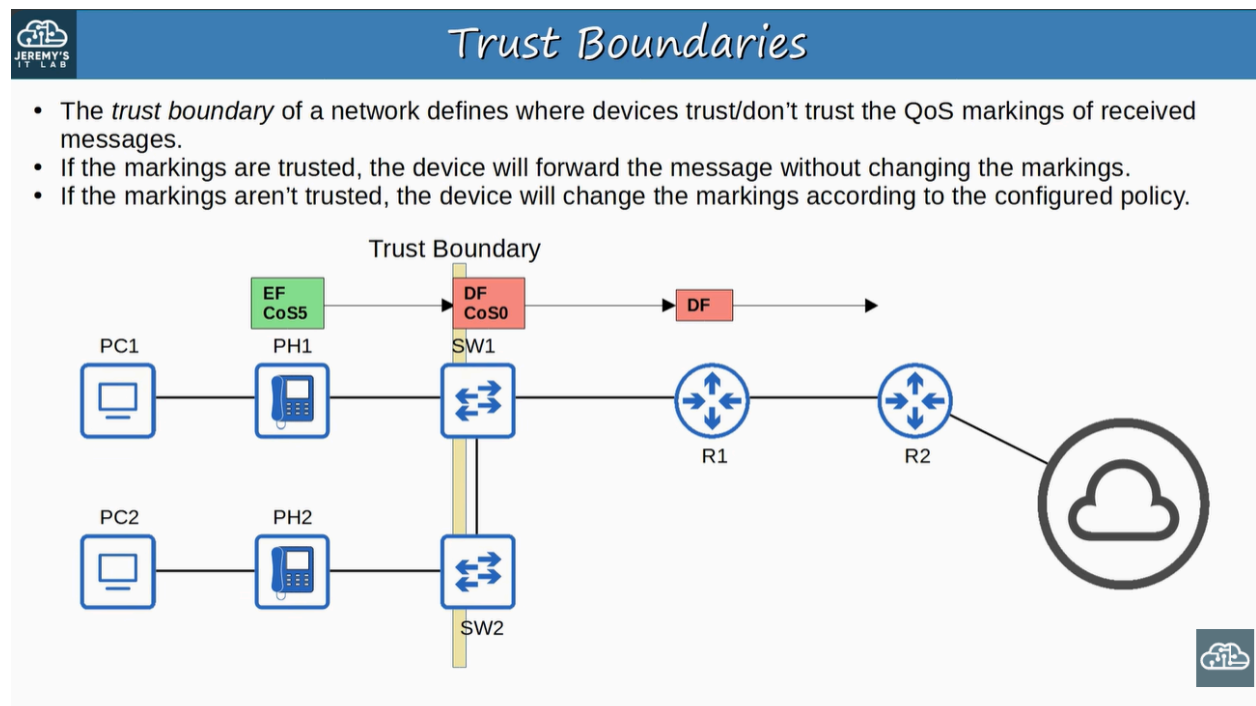
Traffic	Recommended marking
Voice	EF
Interactive video	AF4x
Streaming video	AF3x
High-priority data	AF2x
Best effort	DF

The **x** means the drop-precedence value can vary.

Example:

AF4x → AF41 / AF42 / AF43

10. Trust Boundary



This is very important.

Trust boundary

The point where the network decides:

"Do I trust the QoS marking coming from this device?"

If trusted

The device keeps the marking.

EF/CoS5 → EF/CoS5 → EF/CoS5

If not trusted

The device can **remark** it according to policy.

Example:

User sends:

EF / CoS5

Switch doesn't trust user marking

↓

Changes it to:

DF / CoS0

Why?

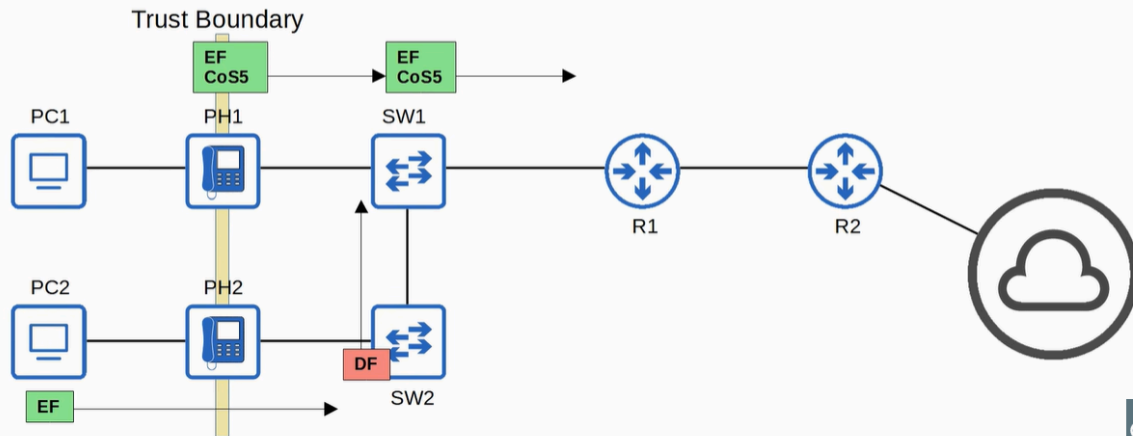
A user should not simply mark their own traffic as:

"I'm extremely important!"

and receive priority.

IP Phone example

- If an IP phone is connected to the switch port, it is recommended to move the trust boundary to the IP phones.
- This is done via configuration on the switch port connected to the IP phone.
- If a user marks their PC's traffic with a high priority, the marking will be changed (not trusted)



IP phones are generally trusted to mark voice traffic.

PC → IP Phone → Switch → Router
 ↑
 Trust boundary

Voice:

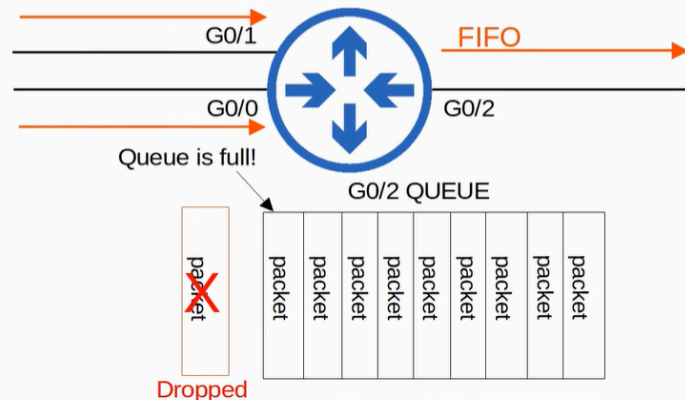
EF / CoS5

PC traffic may be treated differently.

11. Queuing

Queuing/Congestion Management

- When a network device receives traffic at a faster rate than it can forward the traffic out of the appropriate interface, packets are placed in that interface's queue as they wait to be forwarded.
- When the queue becomes full, packets that don't fit in the queue are dropped (tail drop).
- RED and WRED drop packets early to avoid tail drop.



Now we move from:

"What type of packet is this?"

to:

"Where should I wait?"

Why queues?

When traffic arrives faster than the interface can transmit:

Incoming traffic



[QUEUE]



Outgoing interface

Packets wait in a queue.

If the queue becomes full:

New packets may be dropped.

This is called:

Tail Drop

Queue full
↓
New packet
↓
DROP

12. RED / WRED

Instead of waiting until the queue is completely full, we can drop packets **earlier**.

RED

Random Early Detection

Drops packets before the queue becomes completely full.

WRED

Weighted Random Early Detection

Like RED, but can make dropping decisions based on traffic priority/markings.

Main idea

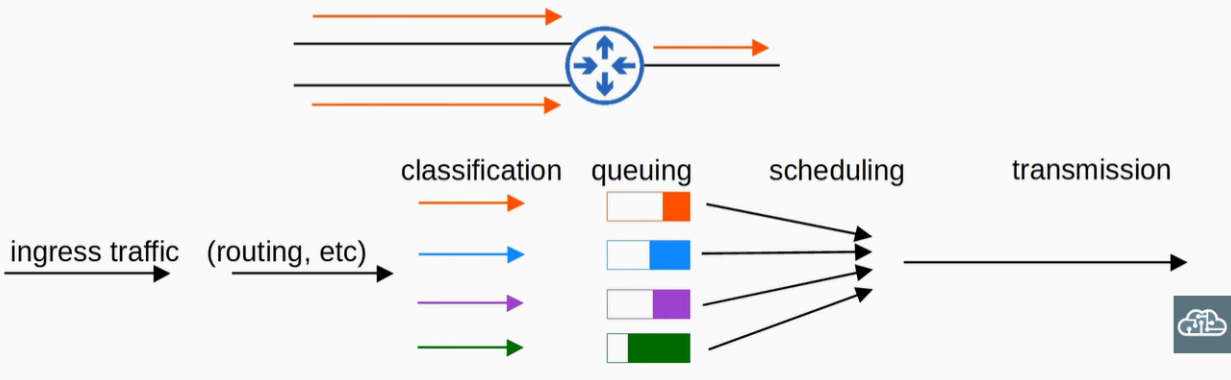
Tail Drop:
Queue FULL → DROP

RED/WRED:
Queue getting FULL → start dropping earlier

13. Multiple Queues

Queuing/Congestion Management

- An essential part of QoS is the use of multiple queues.
 - This is where classification plays a role. The device can match traffic based on various factors (for example the DSCP marking in the IP header) and then place it in the appropriate queue.
- However, the device is only able to forward one frame out of an interface at once, so a *scheduler* is used to decide which queue traffic is forwarded from next.
 - *Prioritization* allows the scheduler to give certain queues more priority than others.



QoS normally uses **multiple queues**.

Classification determines which queue receives the packet.

Example:

Voice —————> Voice Queue
 Video —————> Video Queue
 Web —————> Data Queue
 Backup —————> Low Queue

So the process becomes:

Classification
 ↓
 Queuing
 ↓
 Scheduling
 ↓
 Transmission

14. Scheduling

The router may have several queues, but it can normally transmit only **one frame at a time** on an interface.

So it needs a:

Scheduler

The scheduler decides:

Which queue gets to send next?

15. Round-Robin

The scheduler visits queues one by one.

Example:

Q1 → Q2 → Q3 → Q4

↓

Q1 → Q2 → Q3 → Q4

Each queue gets a turn.

Problem

Voice may have to wait for other queues.

That can create:

- Delay
- Jitter

Therefore, normal round-robin isn't ideal for real-time voice/video.

16. Weighted Round-Robin

Each queue gets a different amount of bandwidth.

Example:

Voice → 60%
Video → 20%
Data → 15%
Background → 5%

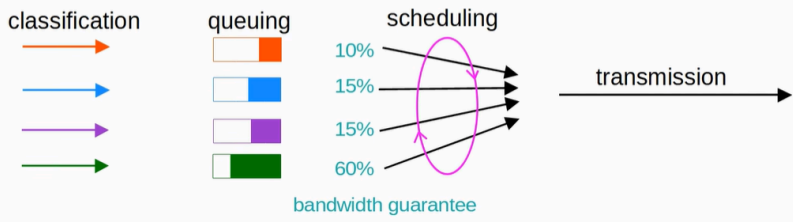
Higher-weight queues get more transmission opportunities.

17. CBWFQ

 JEREMY'S LAB

Queuing/Congestion Management

- A common scheduling method is *weighted round-robin*.
 - **round-robin** = packets are taken from each queue in order, cyclically
 - **weighted** = more data is taken from high priority queues each time the scheduler reaches that queue
- CBWFQ** (Class-Based Weighted Fair Queuing) is a popular method of scheduling, using a weighted round-robin scheduler while guaranteeing each queue a certain percentage of the interface's bandwidth during congestion.
- Round-robin scheduling is not ideal for voice/video traffic. Even if the voice/video traffic receives a guaranteed minimum amount of bandwidth, round-robin can add delay and jitter because even the high priority queues have to wait their turn in the scheduler.



bandwidth guarantee



CBWFQ = Class-Based Weighted Fair Queuing

It uses weighted scheduling and provides each class with a **minimum bandwidth guarantee** during congestion.

Example:

Voice → 10%
Video → 15%
Important → 15%
Data → 60%

Important

CBWFQ gives **bandwidth guarantees**.

But it doesn't necessarily mean:

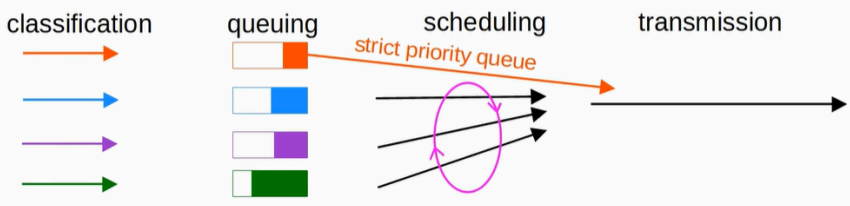
"Voice always goes first."

18. LLQ



Queuing/Congestion Management

- **LLQ** (Low Latency Queuing) designates one (or more) queues as *strict priority queues*.
→ This means that if there is traffic in the queue, the scheduler will always take the next packet from that queue until it is empty.
- This is very effective for reducing the delay and jitter of voice/video traffic.
- However, it has the downside of potentially starving other queues if there is always traffic in the designated strict priority queue.
→ *Policing* (next slide) can control the amount of traffic allowed in the strict priority queue so that it can't take all of the link's bandwidth.



LLQ = Low Latency Queuing

LLQ adds a **strict priority queue**.

Strict priority means:

If the priority queue contains traffic:

Scheduler serves it first.

Example:

Voice Queue ← STRICT PRIORITY

↓
Other Queues

Voice traffic doesn't have to wait for normal round-robin scheduling.

Advantage

Excellent for:

- Voice
- Real-time video

Because it reduces:

- Delay
- Jitter

Problem — Starvation

If the priority queue continuously has traffic:

Voice Voice Voice Voice Voice...

Other queues may not get enough opportunity to transmit.

This is called:

Starvation

That's why LLQ is commonly combined with **policing/rate limits** for the priority traffic.

19. Shaping vs Policing

Shaping and Policing

- Traffic **shaping** and **policing** are both used to control the rate of traffic.
- **Shaping** buffers traffic in a queue if the traffic rate goes over the configured rate.
- **Policing** drops traffic if the traffic rate goes over the configured rate.

*policing also has the option of re-marking the traffic instead of dropping it.
- 'Burst' traffic over the configured rate is allowed for a short period of time.
- This accommodates data applications which typically are 'bursty' in nature. Instead of a constant stream of data, they send data in bursts.
- The amount of burst traffic allowed is configurable.
- In both cases, classification can be used to allow for different rates for different kinds of traffic.
- Why would you want to limit the rate traffic is sent/received?

Both control the **rate of traffic**.

But they handle excess traffic differently.

Traffic Shaping

If traffic exceeds the configured rate:

Buffer it and send later.

Too fast



Buffer



Send at allowed rate

Shaping = Delay

It generally does **not immediately drop** the excess traffic.

Traffic Policing

If traffic exceeds the configured rate:

Drop or remark the excess traffic.

Too fast



Exceeds limit



DROP / REMARK

Policing = Drop/Remark

20. Shaping vs Policing — Easy Table

	Shaping	Policing
Controls rate	✓	✓
Excess traffic	Buffered	Dropped/remarked

Main effect	Delay	Loss/remarking
Usually used	Outbound	Often inbound/outbound
Memory needed	Buffer	Less buffering

Memory trick

Shaping = Slow it down
Policing = Punish excess

21. Burst Traffic

Applications don't always send traffic at a constant rate.

They may send:

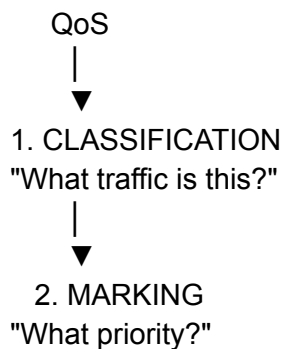


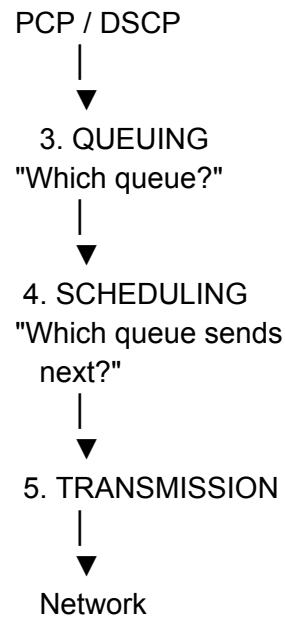
This is called **bursty traffic**.

QoS mechanisms can allow a configurable amount of burst traffic above the normal rate.

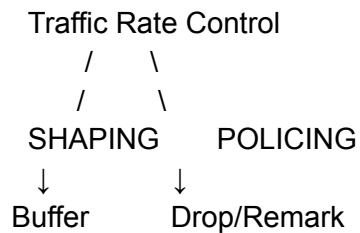
22. Complete QoS Picture

This is the **most important flow to remember**:





And alongside this:



★ Final Exam Revision

Classification

Identify the traffic.

Marking

Label the traffic with priority information.

PCP

Layer 2, 3 bits, 0–7, 802.1Q.

DSCP

Layer 3, 6 bits, 0–63, IP header.

DF

Best effort, DSCP 0.

EF

Low delay/jitter/loss, usually voice, DSCP 46.

AF

4 classes × 3 drop precedences = 12 values.

CS

8 values, backward compatibility with IP Precedence.

Trust Boundary

Decides whether received QoS markings are trusted.

Queuing

Stores packets waiting for transmission.

Tail Drop

Queue full → new packet dropped.

RED/WRED

Drop packets earlier to avoid full queues.

Round-Robin

Queues take turns.

CBWFQ

Weighted queues + bandwidth guarantees.

LLQ

Strict priority queue → excellent for voice/video.

Shaping

Buffer excess traffic and send later.

Policing

Drop or remark excess traffic.

One-line memory chain

Classify → Mark → Queue → Schedule → Transmit

Shaping = Buffer

Policing = Drop/Remark

PCP = L2

DSCP = L3

DF = Best effort

EF = Voice/low latency

AF = Classes + drop precedence

CS = IPP compatibility

This is the clean structure you should keep in your notebook; don't try to memorize all the DSCP/AF values before understanding this flow.